

Dynamic properties of nano-confined water in an ionic liquid

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ABSTRACT

Nano-confined water (a water pocket) was spontaneously formed using an ionic liquid (IL), 1-butyl-3-methylimidazolium nitrate ([C₄mim][NO₃]), in [C₄mim][NO₃]-H₂O and [C₄mim][NO₃]-D₂O mixtures. The static average size of the water pocket in [C₄mim][NO₃]-80 mol% D₂O has been estimated to be around 20 Å, determined via small-angle neutron scattering measurements (Abe et al., *J. Phys. Chem. Lett.*, 2014, **5**, 1175). In this study, the dynamics of the water pocket in the IL were directly examined using incoherent quasielastic neutron scattering. Using the H-D difference of a neutron incoherent scattering cross section, the extent of the diffusion of the water molecules in the water pocket was determined. The water pocket could be characterized due to the slow dynamics of the water compared with those of bulk water.

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1. Introduction

Nano-confined water has been investigated from the viewpoint of the structure and dynamics of water. For instance, water has been artificially confined on the nanoscale using carbon nanotubes [1–7]. A phase transition inside the carbon nanotubes was verified using Raman spectroscopy [1]. Transition temperatures can be drastically changed using different types of carbon nanotubes with different diameters and a variety of water formation was previously observed in a phase diagram [3]. Depending on the carbon nanotube diameter, melting points have been observed to vary. Moreover, using molecular dynamics (MD) simulations, a ring structure of water molecules has been demonstrated in a simulation box [6]. Confined water is also extensively influenced by carbon nanotube walls. In addition to carbon nanotube confinement, water has also been confined inside porous materials [8–13]. Using MD simulations, low density water and high density water have been simulated in nanoporous silica [11]. The nano-confined water was found to have dynamic heterogeneity.

Ionic liquids (ILs), which simply comprise a cation and an anion, possess nanoheterogeneity. Polar and non-polar nano-domains in 1-alkyl-3-methylimidazolium hexafluorophosphate, [C_nmim][PF₆], and 1-alkyl-3-methylimidazolium bis(trifluoromethanesulfonyl)amide, [C_nmim][TFSA] have been theoretically predicted [14]. Here, *n* represents the alkyl chain length of the C_nmim⁺ cation. In theoretical calculations, nanoheterogeneity developed in the simulation box proportionally to the alkyl chain length, *n*. Experimentally, the nanoheterogeneity of [C_nmim][Cl] (*n* = 3, 4, 6, 8, 10) was examined

using X-ray diffraction [15]. This first evidence of nanoheterogeneity provided significant insight into the fundamental science. Related to nanoheterogeneity, the dynamic properties of [C₈mim][PF₆] and [C₈mim][TFSA] were examined by neutron diffraction and neutron spin echo (NSE) techniques [16]. The inter-domain, inter-ionic, and inter-alkyl-chain dynamics was decomposed. The nanoheterogeneity of [C₈mim][NO₃] was simulated not only for the pure ILs, but also for a binary system [17]. Water was found to exist at the nano-domain boundaries in the water-poor region and the nano-confined water (water pocket) formation was observed in the water-rich region. Using small-angle neutron scattering (SANS) [18,19], the water pocket in [C₄mim][NO₃]-D₂O was experimentally probed for 70 < *x* < 90 mol% D₂O. It was clarified that the water pocket is spontaneously formed just upon mixing. Using Raman spectroscopy, at a specific water concentration, it was found that the hydrogen bonding of the water in the water pocket was weakened [20]. Spontaneous formation of the loosely packed water pocket could be explained by the presence of weak hydrogen bonding compared with the strong hydrogen bonding observed in bulk water. Furthermore, at cooling rate of 5 °C/min, no crystallization of [C₄mim][NO₃]-D₂O occurred at 70 < *x* < 94 mol% D₂O [21].

In this study, we investigated the dynamic properties of [C₄mim][NO₃]-80.0 mol% H₂O and [C₄mim][NO₃]-80.2 mol% D₂O using quasielastic neutron scattering (QENS). Using the difference of an incoherent scattering cross section between H and D, a residence time between jumps, τ_0 , of the water in the water pocket was estimated.

2. Experiments

The IL used in this study was [C₄mim][NO₃] (Sigma-Aldrich Co.). Distilled H₂O (99.9% purity, Kanto Chemical Co.) and distilled D₂O (99.9%

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purity, Merck Co.) were used as additives. Due to the IL being hydrophilic, the mixtures were prepared inside a glovebox under a flow of helium gas.

The QENS experiments were carried out using a time-of-flight near backscattering DNA spectrometer at beam line BL02 at the Materials and Life Science Experimental Facility at the Japan Proton Accelerator Research Complex (MLF/J-PARC), Tokai, Ibaraki, Japan, running at 400 kW proton beam power [22–24]. The spectra over an energy transfer range from -0.04 meV to 0.10 meV were measured at an energy resolution of $3.6 \mu\text{eV}$. The momentum transfer, Q , was provided by $4\pi(\sin\theta)/\lambda$ ($\text{\AA}^{-1}$). The Q range was 0.025 – $1.975 \text{ \AA}^{-1}$. Double cylindrical Al cells were used in the QENS experiments. The inner diameter of the outer cylindrical cell (0.25 mm thickness) was 14.0 mm, while the outer diameter of the inner cylindrical cell (0.25 mm thickness) was 13.6 mm. A sample thickness of 0.2 mm was selected to reduce multiple scattering during the experiments. The temperature range was -20 °C to 30 °C, controlled using a top-loading cryo-furnace (Suzuki Shokan Co.). Cooling rate was 1.5 °C/min. The exposure time was about 4–5 h for a single measurement. The spectra obtained were then normalized to those of a vanadium standard. The sample volume of $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.0 mol% H_2O was almost equivalent to that of $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.2 mol% D_2O .

To determine the cooling rate dependence of solidification, simultaneous X-ray diffraction and differential scanning calorimetry (DSC) measurements were carried out using a SmartLab (Rigaku Co.). An incident X-ray wavelength $\lambda = 1.542 \text{ \AA}$ (Cu radiation) was selected. A one dimensional detector (D/tex, Rigaku Co.) was used for a rapid scan.

3. Results and discussion

3.1. Cooling rate dependence of $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.0 mol% H_2O

Fig. 1 depicts the cooling rate dependence of the solidifications of $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.0 mol% H_2O . Upon changing the cooling rate, the X-ray diffraction patterns represented by blue curves were obtained at the minimum temperature (-90 °C). For comparison, X-ray diffraction in the liquid state (25 °C) is depicted in red. At a cooling rate of 1.0 °C/min, crystallization occurred at -39.8 °C (T_C) upon cooling, accompanied by an exothermal peak on the DSC trace. Since the amorphous halo pattern almost diminished below T_C , the mixture was thought to have fully crystallized with a cooling rate of 1.0 °C/min. However, Bragg reflections for ice crystals were not observed. The crystallization of the water pocket was suppressed even by slow cooling. Thus, in $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.0 mol% H_2O , only $[\text{C}_4\text{mim}][\text{NO}_3]$ crystallized, while the water pocket did not crystallize. This is an important observation that crystallization of the nano-confined water pocket is not induced even if the outer IL crystallizes. Upon an increase in the cooling rate, the crystallinity at the minimum temperature proportionally lowered with the cooling rate, where T_C was almost constant. In contrast, a portion of the amorphous halo pattern increased upon an increase in the cooling rate. Finally, above a cooling rate of 4.0 °C/min, crystallization was not observed upon cooling. The amount of crystallization of the IL is therefore heavily dependent on the cooling rate. This is one of the features of the ILs, which causes the nanoheterogeneity in the liquid state and is connected with non-equilibrium behavior. Another feature of the solidification in $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.0 mol% H_2O is that ice crystals are not formed at any cooling rate. From SANS results [18], it was deduced that the size of the nano-confined water pocket is smaller than the critical size required for the nucleation of ice crystals.

3.2. Dynamics of water pocket by QENS

Fig. 2(a) and (b) depicts the incoherent dynamic structure factor, $S_i^{\text{inc}}(Q, \omega)$, of $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.0 mol% H_2O and $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.2 mol% D_2O , at -20 °C upon cooling, respectively. Since the incoherent scattering cross section of hydrogen is much larger than that of other elements,

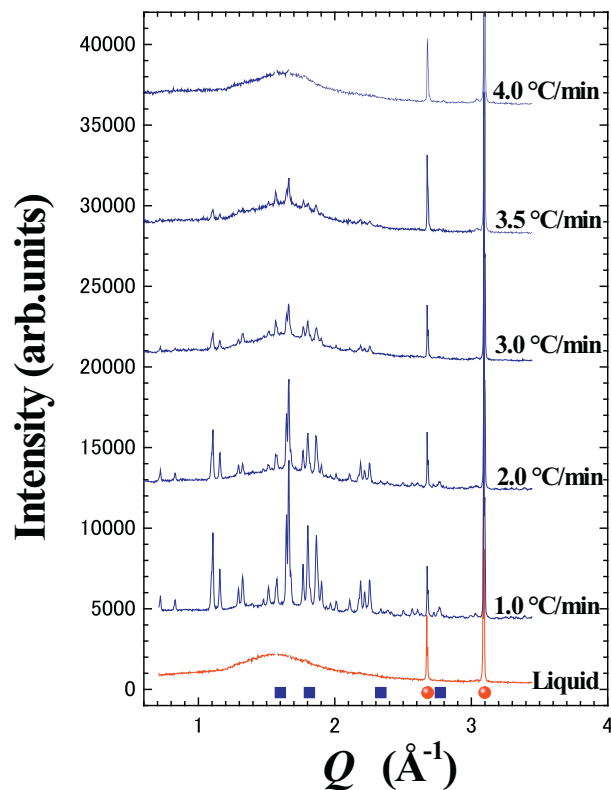


Fig. 1. X-ray diffraction patterns of $[\text{C}_4\text{mim}][\text{NO}_3]$ - 80.0 mol% H_2O . The red curve represents X-ray diffraction in the liquid state at 25 °C. The other curves in blue represent the solid states of the mixture at -90 °C. Upon changing the cooling rates, the crystallinity drastically changed. The red closed circles and blue closed squares correspond to the scattering angles of the Al sample cell and ice crystals, respectively. Obviously, ice crystals were not formed at any of the cooling rates. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the obtained spectra of both samples can be regarded as having incoherent scattering (7 and 12% coherent scattering fractions for $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.0 mol% H_2O and $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.2 mol% D_2O , respectively). In case of $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.0 mol% H_2O , the incoherent dynamic structure factor, $S_{\text{IL}+\text{H}_2\text{O}}^{\text{inc}}(Q, \omega)$, is expressed by [25],

$$S_{\text{IL}+\text{H}_2\text{O}}^{\text{inc}}(Q, \omega) = R(Q, \omega) \otimes [A_{\text{IL}} S_{\text{IL}}^{\text{inc}}(Q, \omega) + A_{\text{H}_2\text{O}} S_{\text{H}_2\text{O}}^{\text{inc}}(Q, \omega)] + BG, \quad (1)$$

where $\otimes$ represents the operator of the convolution. $R(Q, \omega)$ is the instrumental resolution function, which was determined using a vanadium standard sample. $S_{\text{IL}}^{\text{inc}}(Q, \omega)$ and $S_{\text{H}_2\text{O}}^{\text{inc}}(Q, \omega)$ represents the $S^{\text{inc}}(Q, \omega)$ of the ionic liquid and H_2O , respectively. A_{IL} and $A_{\text{H}_2\text{O}}$ are scaling factors. In the case of $[\text{C}_4\text{mim}][\text{NO}_3]$ -80.2 mol% D_2O , Eq. (1) is approximately modified into;

$$S_{\text{IL}+\text{D}_2\text{O}}^{\text{inc}}(Q, \omega) = R(Q, \omega) \otimes [A_{\text{IL}} S_{\text{IL}}^{\text{inc}}(Q, \omega)] + BG. \quad (2)$$

The reason for this is that the incoherent cross section of D is negligibly small. Because of the H-D difference of the incoherent scattering cross section, we can determine the nature of the dynamics of the water in the water pocket by subtraction of $S_{\text{IL}+\text{H}_2\text{O}}^{\text{inc}}(Q, \omega)$ and $S_{\text{IL}+\text{D}_2\text{O}}^{\text{inc}}(Q, \omega)$. The subtraction was conducted to extract the distribution of H_2O in the $\hbar\omega - Q$ plane. If the assumptions described by Eqs. (1) and (2) are correct, smooth distribution could be obtained. After subtraction, in the $\hbar\omega - Q$ plot, Fig. 2(c), a distinct Q dependence of $S_{\text{H}_2\text{O}}^{\text{inc}}(Q, \omega)$ clearly appeared. It should be noticed that the mixture exists in the liquid state at -20 °C ($>T_C$). The decomposed $S_{\text{H}_2\text{O}}^{\text{inc}}(Q, \omega)$ contains dynamic information about the self-diffusion of the water molecule in

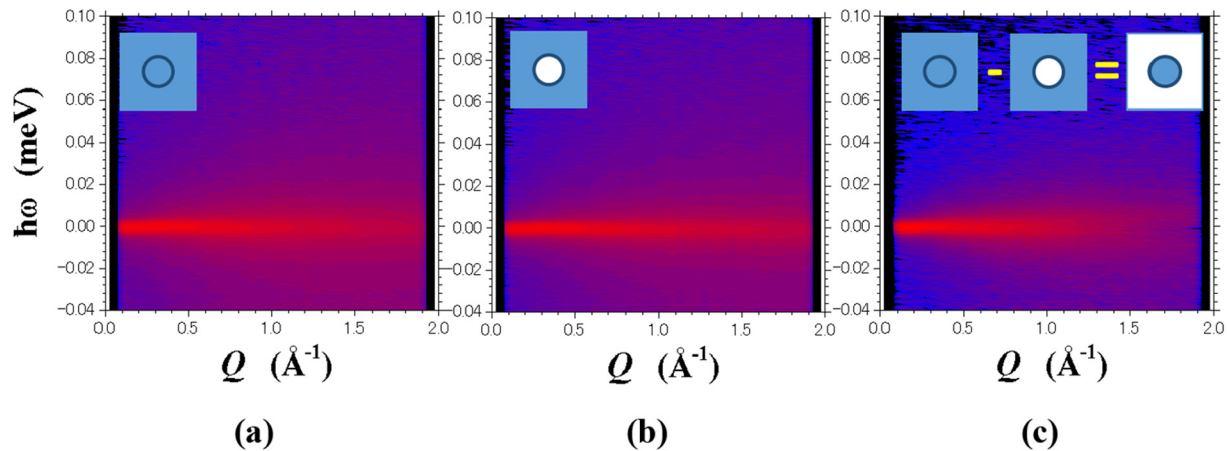


Fig. 2. The dynamic structure factors, $S^{\text{inc}}(Q, \omega)$, of (a) [C₄mim][NO₃]-80.0 mol% H₂O, (b) [C₄mim][NO₃]-80.2 mol% D₂O, and (c) the difference between the H₂O- and D₂O-based mixtures. The intensity is plotted by a logarithmic scale.

the water pocket. For further analysis, the observed energy spectra of [C₄mim][NO₃]-80.0 mol% H₂O were fitted using the following equation:

$$S^{\text{inc}}_{\text{IL+H}_2\text{O}}(Q, \omega) = R(Q, \omega) \otimes [A_{\text{IL}_1} \mathcal{F}[f(t)] + A_{\text{IL}_2} L_{\text{IL}_2}(\Gamma_{\text{IL}_2}, \omega) + A_{\text{H}_2\text{O}} L_{\text{H}_2\text{O}}(\Gamma_{\text{H}_2\text{O}}, \omega)] + BG \quad (3)$$

where $L_i(\Gamma_i, \omega)$, A_i and Γ_i represent the Lorentzian function, the intensity of the Lorentzian function and the half width at half maximum (HWHM) of the Lorentzian function of the i -component, respectively. $f(t)$ denotes the Kohlrausch-Williams-Watts (KWW) function [25–27]. The KWW function is expressed as;

$$f(t) = \exp\left\{-\left(\frac{t}{\tau_R}\right)^\beta\right\} \quad (4)$$

β represents the stretching exponential and is restricted to between 0 and 1. τ_R is the characteristic relaxation time. $\mathcal{F}$ represents the Fourier transform. At first, the QENS profiles of [C₄mim][NO₃]-80.2 mol% D₂O at -20°C were fitted to obtain the parameters of the first and second terms of Eq. (3), as depicted in Fig. 3(a), because the third term of Eq. (3) is negligible due to the small incoherent scattering cross section of D. Then, the QENS profiles of [C₄mim][NO₃]-80.0 mol% H₂O were fitted using the fixed parameters obtained using the D₂O-based mixtures as depicted in Fig. 3(b). The fixing parameters are equivalent to subtract the D₂O component from the total scattering. Finally, the parameters derived only from water in the water pocket were determined. Fig. 4 depicts the Q^2 dependence of $\Gamma_{\text{H}_2\text{O}}(Q)$ of water in the water pocket. In a jump diffusion model [25–28], $\Gamma(Q)$ is provided by;

$$\Gamma_{\text{H}_2\text{O}}(Q) = \frac{D_{\text{H}_2\text{O}} Q^2}{1 + D_{\text{H}_2\text{O}} Q^2 \tau_0} \quad (5)$$

where $D_{\text{H}_2\text{O}}$ and τ_0 correspond to the diffusion coefficient of water and mean residence time of water, respectively. The solid line in Fig. 4 was obtained by jump diffusion model using Eq. (5). $D_{\text{H}_2\text{O}}$ was found to be $1.8 \times 10^{10} \text{ \AA}^2 \text{ s}^{-1}$, and τ_0 was calculated to be 20. ps. For comparison, the $\Gamma_{\text{bulk}}(Q)$ value of bulk water [29] is represented by the broken line in Fig. 4. Since the D_{bulk} and τ_0 values of bulk water were found to be $4.2 \times 10^{10} \text{ \AA}^2 \text{ s}^{-1}$ and 22.7 ps [29], the water molecules confined in the water pocket displayed slower dynamics above T_C , than those of bulk water. The QENS results are consistent with previous QENS studies [30,31] and those of Raman spectroscopy [20]. With decreasing the nano-pore size, translational diffusion of the confined water became smaller, since growth of the hydrogen-bond network of water is

suppressed. In the Raman spectra of the stretching mode of water in the [C₄mim][NO₃] $\times$ mol% D₂O, the Raman intensity of the strong hydrogen bonding of the water pocket (70–90 mol%) became weaker than that of pure water. In the MD simulations of the confined water inside carbon nanotube [4], relative amount of hydrogen bonds decreased drastically below 20 $\text{\AA}$ in the simulation box. Here it is confirmed that, in the confined water on the nanoscale, not only can a static average size be determined by SANS but also slow dynamics (weak hydrogen

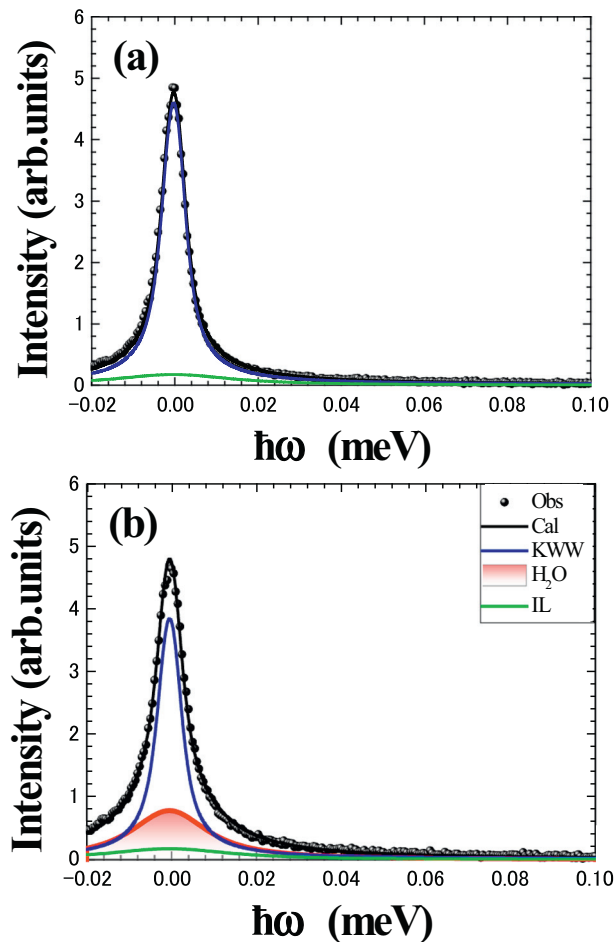


Fig. 3. QENS spectra and fitted results of (a) [C₄mim][NO₃]-80.2 mol% D₂O and (b) [C₄mim][NO₃]-80.0 mol% H₂O at -20°C . Q is fixed at $1.18 \text{ \AA}^{-1}$.

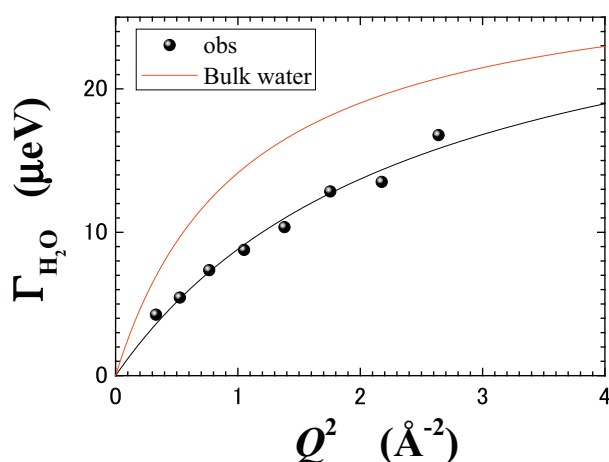


Fig. 4. Q^2 dependence of HWHM, $\Gamma_{\text{H}_2\text{O}}(Q)$, at -20°C . The solid curve reveals that the fitted values are based on a jump diffusion model (see text for details). The broken line represents the $\Gamma_{\text{bulk}}(Q)$ of bulk water at -20°C for comparison.

bonding) are observed to contribute to a disturbing of the nucleation of ice crystal at a low temperature. This is a significant finding of the spontaneous formation of a water pocket on the nanoscale, with the loosely packed water pocket being characterized by slow dynamics compared with those of bulk water.

4. Conclusions

The dynamical properties of water in a water pocket formed by an IL in different $\text{H}_2\text{O}/\text{D}_2\text{O}$ mixtures were determined using incoherent QENS. The H-D difference of an incoherent scattering cross section provides evidence of the slow dynamics of the nano-confined water in the water pocket. The slow dynamics are in agreement with the weak hydrogen bonding observed in Raman spectra. Spontaneous formation and no crystallization of the water pocket in the IL originate not only from the static average size but also from the slow dynamics of the nano-confined water.

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References

- [1] K.V. Agrawal, S. Shimizu, L.W. Drahushuk, D. Kilcoyne, M.S. Strano, Observation of extreme phase transition temperatures of water confined inside isolated carbon nanotubes, *Nat. Nanotechnol.* 12 (2017) 267–273.
- [2] Y. Maniwa, H. Kataura, M. Abe, S. Suzuki, Y. Achiba, H. Kira, K. Matsuda, Phase transition in confined water inside carbon nanotubes, *J. Phys. Soc. Jpn.* 71 (2002) 2863–2866.
- [3] D. Takaiwa, I. Hatano, K. Koga, H. Tanaka, Phase diagram of water in carbon nanotubes, *Proc. Natl. Acad. Sci.* 105 (2008) 39–43.
- [4] H. Ye, H. Zhang, Z. Zhang, Y. Zheng, Size and temperature effects on the viscosity of water inside carbon nanotubes, *Nanoscale Res. Lett.* 6 (2011) 87–5.
- [5] S.O. Diallo, M. Jażdżewska, J.C. Palmer, E. Mamontov, K.E. Gubbins, M. Sliwińska-Bartkowiak, Dynamics of nanoconfined water under pressure, *Phys. Rev. E* 88 (2013) 022316–6.
- [6] G.J. Wang, N.G. Hadjiconstantinou, Why are fluid densities so low in carbon nanotubes? *Phys. Fluids* 27 (2015) 052006–11.
- [7] X. Ma, S. Cambré, W. Wenseleers, S.K. Doorn, H. Htoon, Quasiphase transition in a single file of water molecules encapsulated in (6,5) carbon nanotubes observed by temperature-dependent photoluminescence spectroscopy, *Phys. Rev. Lett.* 118 (2017) 027402–7.

- [8] F.-X. Coudert, F. Cailliez, R. Vuilleumier, A.H. Fuchs, A. Boutin, Water nanodroplets confined in zeolite pores, *Faraday Discuss.* 141 (2009) 377–398.
- [9] D. Laage, W.H. Thompson, Reorientation dynamics of nanoconfined water: power-law decay, hydrogen-bond jumps, and test of a two-state model, *J. Chem. Phys.* 136 (2012) 044513–7.
- [10] G.K. Lockwood, S.H. Garofalini, Proton dynamics at the water–silica Interface via dissociative molecular dynamics, *J. Phys. Chem. C* 118 (2014) 29750–29759.
- [11] A. Shekhar, R.K. Kalia, A. Nakano, P. Vashishta, C.K. Alm, A. Malthe-Sørensen, Universal stretched exponential relaxation in nanoconfined water, *Appl. Phys. Lett.* 105 (2014) 161907–4.
- [12] G. Goracci, M. Monasterio, H. Jansson, S. Cerveny, Dynamics of nano-confined water in Portland cement – comparison with synthetic C-S-H gel and other silicate materials, *Sci. Rep.* 7 (2017) 8258–10.
- [13] B. Rossi, V. Venuti, A. Mele, C. Punta, L. Melone, de F. D'Amico, A. Gessini, V. Crupi, D. Majolino, F. Trotta, C. Masciovecchio, Vibrational signatures of the water behaviour upon confinement in nanoporous hydrogels, *Phys. Chem. Chem. Phys.* 18 (2016) 12252–12259.
- [14] J.N.A. Canongia Lopes, A.A.H. Pádua, Nanostructural organization in ionic liquids, *J. Phys. Chem. B* 110 (2006) 3330–3335.
- [15] A. Triolo, O. Russina, H.-J. Bleif, E. Di Cola, Nanoscale segregation in room temperature ionic liquids, *J. Phys. Chem. B* 111 (2007) 4641–4644.
- [16] M. Kofu, M. Nagao, T. Ueki, Y. Kitazawa, Y. Nakamura, S. Sawamura, O. Yamamuro, Heterogeneous slow dynamics of imidazolium-based ionic liquids studied by neutron spin echo, *J. Phys. Chem. B* 117 (2013) 2773–2781.
- [17] W. Jiang, Y. Wang, G.A. Voth, Molecular dynamics simulation of nanostructural organization in ionic liquid/water mixtures, *J. Phys. Chem. B* 111 (2007) 4812–4818.
- [18] H. Abe, T. Takekiyo, M. Shigemitsu, Y. Yoshimura, S. Tsuge, T. Hanasaki, K. Ohishi, S. Takata, J. Suzuki, Direct evidence of confined water in room-temperature ionic liquids by complementary use of small-angle X-ray and neutron scattering, *J. Phys. Chem. Lett.* 5 (2014) 1175–1180.
- [19] H. Abe, T. Takekiyo, M. Shigemitsu, Y. Yoshimura, S. Tsuge, T. Hanasaki, K. Ohishi, S. Takata, J. Suzuki, Size-tunable confined water in a room temperature ionic liquid, *JPS Conf. Proc.* 8 (2015) 033001–6.
- [20] H. Abe, T. Takekiyo, M. Aono, H. Kishimura, Y. Yoshimura, N. Hamaya, Polymorphs in room-temperature ionic liquids: hierarchical structure, confined water and pressure-induced frustration, *J. Mol. Liq.* 210 (2015) 200–214.
- [21] H. Abe, T. Takekiyo, Y. Yoshimura, K. Saihara, A. Shimizu, Anomalous freezing of Nano-confined water in room-temperature ionic liquid 1-butyl-3-methylimidazolium nitrate, *ChemPhysChem* 17 (2016) 1136–1142.
- [22] K. Shibata, N. Takahashi, Y. Kawakita, M. Matsuura, T. Yamada, T. Tomimaga, W. Kambara, M. Kobayashi, Y. Inamura, T. Nakatani, K. Nakajima, M. Arai, The performance of TOF near backscattering spectrometer DNA in MLF, J-PARC, *JPS Conf. Proc.* 8 (2015) 036022–7.
- [23] H. Seto, S. Itoh, T. Yokoo, H. Endo, K. Nakajima, K. Shibata, R. Kajimoto, S. Ohira-Kawamura, M. Nakamura, Y. Kawakita, H. Nakagawa, T. Yamada, Inelastic and quasi-elastic neutron scattering spectrometers in J-PARC, *Biochim. Biophys. Acta* 1861 (2017) 3651–3660.
- [24] K. Nakajima, Y. Kawakita, S. Itoh, J. Abe, K. Aizawa, H. Aoki, H. Endo, M. Fujita, K. Funakoshi, W. Gong, M. Harada, S. Harjo, T. Hattori, M. Hino, T. Honda, A. Hoshikawa, K. Ikeda, T. Ino, T. Ishigaki, Y. Ishikawa, H. Iwase, T. Kai, R. Kajimoto, T. Kamiyama, N. Kaneko, D. Kawana, S. Ohira-Kawamura, T. Kawasaki, A. Kimura, R. Kiyanagi, K. Kojima, K. Kusaka, S. Lee, S. Machida, T. Masuda, K. Mishima, K. Mitamura, M. Nakamura, S. Nakamura, A. Nakao, T. Oda, T. Ohhara, K. Ohishi, H. Ohshita, K. Oikawa, T. Otomo, A. Sano-Furukawa, K. Shibata, T. Shinohara, K. Soyama, J. Suzuki, K. Suzuya, A. Takahara, S. Takata, M. Takeda, Y. Toh, S. Torii, N. Torikai, N.L. Yamada, T. Yamada, D. Yamazaki, T. Yokoo, M. Yonemura, Hideki Yoshizawa, Materials and life science experimental facility (MLF) at the Japan Proton Accelerator Research Complex II: neutron scattering instruments, *Quantum Beam Sci.* 1 (2017) 9–59.
- [25] D.L. Price, O. Borodin, M.A. González, M. Kofu, K. Shibata, T. Yamada, O. Yamamuro, M.-L. Saboungi, Relaxation in a prototype ionic liquid: influence of water on the dynamics, *J. Phys. Chem. Lett.* 8 (2017) 715–719.
- [26] J.H. Wu, Q. Jia, The heterogeneous energy landscape expression of KWW relaxation, *Sci. Rep.* 6 (2016) 20506–20509.
- [27] P. Le, E. Fratini, L. Zhang, K. Ito, E. Mamontov, P. Baglioni, S.-H. Chen, Quasi-elastic neutron scattering study of hydration water in synthetic cement: an improved analysis method based on a new global model, *J. Phys. Chem. C* 121 (2017) 12826–12833.
- [28] G. Briganti, G. Rogati, A. Parmentier, M. Maccarini, F. De Luca, Neutron scattering observation of quasi-free rotations of water confined in carbon nanotubes, *Sci. Rep.* 7 (2017) 45021–10.
- [29] J. Teixeira, M.-C. Bellissent-Funel, S.H. Chen, A.J. Dianoux, Experimental determination of the nature of diffusive motions of water molecules at low temperatures, *Phys. Rev. A* 31 (1985) 1913–1917.
- [30] S. Takahara, M. Nakano, S. Kittaka, Y. Kuroda, T. Mori, H. Hamano, T. Yamaguchi, Neutron scattering study on dynamics of water molecules in MCM-41, *J. Phys. Chem. B* 103 (1999) 5814–5819.
- [31] S. Takahara, N. Sumiyama, S. Kittaka, T. Yamaguchi, M.-C. Bellissent-Funel, Neutron scattering study on dynamics of water molecules in MCM-41. 2. determination of translational diffusion coefficient, *J. Phys. Chem. B* 109 (2005) 11231–11239.